Interband Absorption in Gold from Fermi's Golden Rule: a first-principles rederivation of Rosei's ε_2 at the X and L critical points, with the photonic density of states

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Abstract

This document rederives, from first principles and with every prefactor tracked, the Rosei-model interband contribution to the imaginary part of the dielectric function of gold, $\varepsilon_2(\hbar\omega, T)$, at the X and L critical points [Guerrisi, Rosei & Winsemius (GRW), Phys. Rev. B **12**, 557 (1975)]. The route is the one developed in my own notes: Fermi's Golden Rule (FGR) for Bloch states in **k**-space, reduction of the six-dimensional transition sum to a one-dimensional energy integral through the linearization $(u, v) = (k_\perp^2, k_\parallel^2)$, and resolution of the double δ-constraint by a single 2×2 Jacobian. The two critical points are treated strictly according to their band topology; where the derivations diverge, the two paths are displayed side by side. The known transcription errors in GRW's printed Eqs. (6) and (8) are corrected, and a genuine ambiguity in the assignment of the parallel effective masses at X — which decides between a closed transition-energy window and GRW's open, sub-gap-tailed one — is resolved into two clearly separated branches, together with the observation that the above-gap line *shape* is blind to the choice. Finally, the photonic density of states is derived by field quantization, the spontaneous-emission counterpart of the absorption integral is assembled, and the pair is shown to close exactly into the van Roosbroeck–Shockley/Kirchhoff relation, providing a global consistency check of all prefactors.

Contents

1 Scope, sources and reading map

Three families of documents about interband absorption coexist in this project:

1. **My derivation chain** (appendices A1, A8, A8.L and the notes *Absorption Integral*, *Interband Absorption at X and L*, *Equivalence of My Absorption Integral and Rosei's*): FGR in **k**-space, the (u, v) linearization, and the verified Jacobian bridge to Rosei's formulas.

2. **Attempts at recreating Rosei's own derivation** (`interband_derivation.tex`, v2, `rosei_equivalence_derivation.tex`); these are AI-drafted, mutually inconsistent in their band-sign conventions, and are superseded by the present document.

3. **The primary source**, GRW 1975, whose printed Eqs. (6) and (8) contain transcription defects (App. A).

This document keeps only what survives verification. Section 3 builds the golden-rule machinery with all constants; Sec. 4 performs the **k**-space reduction in a form common to both critical points; Sec. 5 splits into the X and L columns; Sec. 6 assembles ε_2; Sec. 7 derives the photonic density of states and the emission spectrum. Appendices collect the GRW corrections, a dictionary of the conventions used across the repository, and a full dimensional audit.

2 Conventions

Units	SI throughout; Gaussian equivalents given where GRW is quoted.
Energy zero	Fermi level, $E_F = 0$; all electron energies E from E_F.
Electron charge	$-e$, $e > 0$; electron (bare) mass m.
Spin	carried explicitly as a factor 2, never absorbed silently.
Local axes	at each critical point, $k_\parallel$ along the ΓX (resp. ΓL) axis, $k_\perp$ radial in the perpendicular plane; **k** measured from the critical point.
Band coefficients	$A_i = \hbar^2/2m_{i\perp}$, $B_i = \hbar^2/2\lvert m_{i\parallel}\rvert$, $i = c, v$; all $A_i, B_i > 0$, signs of the dispersions carried by explicit $\pm$ and by $\sigma_c = \pm 1$ (Sec. 4.1).
Occupation	$f(E) = \left[e^{E/k_BT} + 1\right]^{-1}$ (equilibrium; non-equilibrium f^S, f^P substitute).
Photon	energy $\hbar\omega$, wavevector **q**, polarization $\hat{\boldsymbol{\varepsilon}}_\lambda$.

3 Fermi's Golden Rule in k-space

3.1 Minimal coupling

With the radiation field described by a vector potential $\mathbf{A}$ in the Coulomb gauge ($\nabla\cdot\mathbf{A}=0$), the one-electron Hamiltonian is

$$\hat{H}=\frac{[\hat{\mathbf{p}}+e\mathbf{A}(\hat{\mathbf{r}},t)]^2}{2m}+V(\hat{\mathbf{r}})=\underbrace{\frac{\hat{p}^2}{2m}+V}_{\hat{H}_0}+\underbrace{\frac{e}{2m}\big(\mathbf{A}\cdot\hat{\mathbf{p}}+\hat{\mathbf{p}}\cdot\mathbf{A}\big)}_{\hat{H}_{\text{int}}}+\underbrace{\frac{e^2A^2}{2m}}_{\text{dropped}}. \tag{3.1}$$

In the Coulomb gauge $\hat{\mathbf{p}}\cdot\mathbf{A}=\mathbf{A}\cdot\hat{\mathbf{p}}$, so

$$\boxed{\hat{H}_{\text{int}}=\frac{e}{m}\,\mathbf{A}\cdot\hat{\mathbf{p}}} \tag{3.2}$$

(the factor $\frac{1}{2}$ of the symmetrized form is consumed by the identity of the two terms; note A1 drops the $\frac{1}{2}$ one line early but lands on the correct (3.2)). The A^2 term does not connect different bands to first order and is dropped.

3.2 Quantized field: absorption, stimulated and spontaneous emission

Expanding the field in a quantization volume V,

$$\mathbf{A}(\mathbf{r})=\sum_{\mathbf{q},\lambda}\sqrt{\frac{\hbar}{2\varepsilon_0\omega_qV}}\,\hat{\boldsymbol{\varepsilon}}_\lambda\Big(\hat{a}_{\mathbf{q}\lambda}\,\mathrm{e}^{i\mathbf{q}\cdot\mathbf{r}}+\hat{a}^\dagger_{\mathbf{q}\lambda}\,\mathrm{e}^{-i\mathbf{q}\cdot\mathbf{r}}\Big), \tag{3.3}$$

the matrix elements of $\hat{H}_{\text{int}}$ between joint electron–photon states carry the boson factors

$$\big|\langle n_q-1|\hat{a}|n_q\rangle\big|^2=n_q\quad\text{(absorption)},\qquad \big|\langle n_q+1|\hat{a}^\dagger|n_q\rangle\big|^2=n_q+1\quad\text{(emission)}. \tag{3.4}$$

The "+1" is the spontaneous channel: it survives at $n_q=0$ and is invisible to a purely classical $\mathbf{A}$. This answers the footnote question left open in note A1: the semiclassical golden rule gives absorption and *stimulated* emission only; spontaneous emission requires (3.4), and its mode sum is what produces the photonic density of states in Sec. 7.

3.3 Bloch states and vertical transitions

For Bloch states $\psi_{n\mathbf{k}}=\mathrm{e}^{i\mathbf{k}\cdot\mathbf{r}}u_{n\mathbf{k}}(\mathbf{r})/\sqrt{V}$, the matrix element of $\mathrm{e}^{\pm i\mathbf{q}\cdot\mathbf{r}}\,\hat{\boldsymbol{\varepsilon}}\hat{\mathbf{p}}$ enforces crystal-momentum conservation $\mathbf{k}'=\mathbf{k}\pm\mathbf{q}$. At optical frequencies $q\sim\omega/c\sim10^{-3}\,\text{Å}^{-1}$ is negligible on the scale of the Brillouin zone, so transitions are vertical ($\mathbf{k}'=\mathbf{k}$, dipole approximation) and

$$M_{cv}(\mathbf{k})\equiv\langle c\,\mathbf{k}|\,\hat{\boldsymbol{\varepsilon}}\cdot\hat{\mathbf{p}}\,|v\,\mathbf{k}\rangle,\qquad \big|\langle f|\hat{H}_{\text{int}}|i\rangle\big|^2=\Big(\frac{e}{m}\Big)^2\frac{\hbar}{2\varepsilon_0\omega V}\,|M_{cv}(\mathbf{k})|^2\times\{n_q\text{ or }n_q+1\}. \tag{3.5}$$

3.4 Golden-rule rates and ε_2

FGR for one photon mode, then summing over $\mathbf{k}$ (factor 2 for spin, $\sum_{\mathbf{k}}\to V\int\mathrm{d}^3k/(2\pi)^3$) with Pauli blocking, gives the net photon absorption rate. Equating the absorbed power to the electromagnetic dissipation rate $P_{\text{abs}}=\frac{1}{2}\varepsilon_0\,\varepsilon_2(\omega)\,\omega|\mathbf{E}_0|^2V$ of a classical field $\mathbf{E}=\mathbf{E}_0\cos\omega t$ (equivalently $A_0=E_0/\omega$) yields the standard exact result

$$\boxed{\varepsilon_2(\omega)=\frac{\pi e^2}{\varepsilon_0m^2\omega^2}\,\frac{2}{(2\pi)^3}\int_{\text{BZ}}\mathrm{d}^3k\,\big|M_{cv}(\mathbf{k})\big|^2\,\delta\big(E_c(\mathbf{k})-E_v(\mathbf{k})-\hbar\omega\big)\,\big[f(E_v)-f(E_c)\big]} \tag{3.6}$$

per pair of bands (Gaussian form: replace $e^2/\varepsilon_0 \to 4\pi e_G^2$, i.e. prefactor $4\pi^2 e_G^2/m^2\omega^2$). The factor $[f(E_v) - f(E_c)]$ is the *net* of absorption minus stimulated emission; GRW's $[1 - f(E)]$ is its limit for a fully occupied lower band, cf. Sec. 4.4.

Near a critical point the interband momentum matrix element varies slowly; following Rosei we freeze it and average over polarization,

$$|M_{cv}|^2 \;\longrightarrow\; \overline{|\hat{\varepsilon}\cdot\mathbf{p}_{cv}|^2} = \tfrac{1}{3}|P|^2, \qquad |P|^2 \equiv |\mathbf{p}_{cv}|^2 \;\; (\text{SI momentum}^2). \tag{3.7}$$

4 Reduction of the k-integral: the common machinery

4.1 The Rosei two-band model

Both critical points of gold are described by axially symmetric parabolic pairs. With the conventions of Sec. 2 and the *topology switch* σ_c,

$$\boxed{\begin{aligned} E_c(\mathbf{k}) &= E_{0c} + A_c k_\perp^2 + \sigma_c B_c k_\parallel^2, & \sigma_c^X &= -1 \;\; (\text{conduction saddle}),\\ E_v(\mathbf{k}) &= -\hbar\omega_v - A_v k_\perp^2 - B_v k_\parallel^2, & \sigma_c^L &= +1 \;\; (\text{conduction minimum}),\end{aligned}} \tag{4.1}$$

where $E_{0c} > 0$ is the empty level at the critical point ($\hbar\omega_{X_6^-}$ or $\hbar\omega_{L_6^-}$), $\hbar\omega_v > 0$ the occupied d level below E_F ($\hbar\omega_{X_7^+}$ or $\hbar\omega_{L^+}$), and the valence band is a local maximum in both directions at both points (all four C&S curvatures negative). The vertical transition energy is

$$\Omega(\mathbf{k}) \equiv E_c - E_v = \mathcal{E}_g + \bar{A}\,k_\perp^2 + \bar{B}\,k_\parallel^2, \qquad \mathcal{E}_g = E_{0c} + \hbar\omega_v, \quad \bar{A} = A_c + A_v, \quad \boxed{\bar{B} = B_v + \sigma_c B_c} \tag{4.2}$$

— the entire topological difference between X and L is carried by σ_c through $\bar{B}$ (and through the determinant $\mathcal{D}$ below).

Gold parameters (levels from GRW's fit; masses in units of m, extracted from Christensen & Seraphin 1971, Fig. 5 — see the branch discussion in Sec. 5.1 for the starred entries):

	$\mathcal{E}_g$ (eV)	E_{0c} (eV)	$\hbar\omega_v$ (eV)	$m_{c\perp}$	$m_{c\parallel}$	$m_{v\perp}$	$m_{v\parallel}$	N
X	1.94	0.17	1.77	0.31	-0.40^*	-0.19	-0.15^*	3 (=6 halves)
L	2.45	0.89	1.56	0.24	$+0.12$	-0.70	-1.03	4 (=8 halves)

N counts *full* inequivalent critical points; GRW count half-neighborhoods (6 X-faces, 8 L-faces), which is equivalent because the reduction below integrates over the full $\pm k_\parallel$ neighborhood of each of the N points. Getting this factor wrong is a silent factor-2 error; here it is pinned explicitly.

4.2 Linearization $(u, v) = (k_\perp^2, k_\parallel^2)$

This is the step where my derivation departs from (and simplifies) Rosei's geometric construction. On the quadrant $u, v \geq 0$ the two constraints of (3.6) — fixing the final-state energy E and the transition energy $\Omega = \hbar\omega$ — are *linear*:

$$\begin{pmatrix} E - E_{0c} \\ \hbar\omega - \mathcal{E}_g \end{pmatrix} = \underbrace{\begin{pmatrix} A_c & \sigma_c B_c \\ \bar{A} & \bar{B} \end{pmatrix}}_{M} \begin{pmatrix} u \\ v \end{pmatrix}, \qquad \mathcal{D} \equiv \det M = A_c B_v - \sigma_c\, A_v B_c. \tag{4.3}$$

Hence

$$\boxed{u = \frac{\bar{B}\,(E - E_{0c}) - \sigma_c B_c(\hbar\omega - \mathcal{E}_g)}{\mathcal{D}}, \qquad v = k_\parallel^2 = \frac{A_c(\hbar\omega - \mathcal{E}_g) - \bar{A}\,(E - E_{0c})}{\mathcal{D}}} \tag{4.4}$$

and the physical window is exactly $\{u \geq 0\} \cap \{v \geq 0\}$. Note the two point-specific determinants:

$$\mathcal{D}_X = A_c B_v + A_v B_c > 0 \;\; \text{always}, \qquad \mathcal{D}_L = A_c B_v - A_v B_c \;\; (\text{either sign; } < 0 \text{ for Au}). \tag{4.5}$$

4.3 Resolving the double δ: the kernel $K(E, \hbar\omega)$

Define the double-constrained $\mathbf{k}$-space measure (per critical point, full $\pm k_\parallel$ neighborhood, no spin):

$$K(E,\hbar\omega) \equiv \int \mathrm{d}^3k\; \delta\big(E_c(\mathbf{k}) - E\big)\, \delta\big(\Omega(\mathbf{k}) - \hbar\omega\big). \tag{4.6}$$

With axial symmetry, $\mathrm{d}^3k = 2\pi k_\perp\, \mathrm{d}k_\perp\, \mathrm{d}k_\parallel$ over $k_\parallel \in (-\infty, \infty)$; substituting $\mathrm{d}u = 2k_\perp \mathrm{d}k_\perp$ and, on each half-line, $\mathrm{d}v = 2k_\parallel \mathrm{d}k_\parallel$,

$$\int \mathrm{d}^3k = 2\pi \cdot \tfrac{1}{2} \int_0^\infty \mathrm{d}u \cdot 2 \cdot \tfrac{1}{2} \int_0^\infty \frac{\mathrm{d}v}{\sqrt{v}} = \pi \int_0^\infty \mathrm{d}u \int_0^\infty \frac{\mathrm{d}v}{\sqrt{v}}, \tag{4.7}$$

where the explicit "2" counts the two hyperbola branches $\pm k_\parallel$ (this factor was implicit — and halved — in the *Equivalence* note, which parameterized only $k_\parallel > 0$; the convention here matches the N counting of Sec. 4.1). The linear map (4.3) collapses both δ's with Jacobian $1/|\mathcal{D}|$:

$$\boxed{K(E,\hbar\omega) = \frac{\pi}{|\mathcal{D}|\, k_\parallel(E,\hbar\omega)} = \frac{4\pi\,\mathcal{F}^2}{\hbar^4\, k_\parallel(E,\hbar\omega)}, \qquad \mathcal{F} \equiv \frac{\hbar^2}{2\sqrt{|\mathcal{D}|}}} \tag{4.8}$$

supported on the window $u, v \ge 0$, with $k_\parallel = \sqrt{v}$ from (4.4). Equivalently, in the $(k_\perp, k_\parallel)$ variables the same result follows from $|\det J| = 4k_\perp k_\parallel |\mathcal{D}|$ with $J = \partial(E_c, \Omega)/\partial(k_\perp, k_\parallel)$ — the form verified in the *Equivalence* note.

The mass form of $\mathcal{F}$ reproduces GRW Eq. (5) at X and its L analogue:

$$\mathcal{F}_X = \left[\frac{m_{v\perp}m_{c\parallel} + m_{v\parallel}m_{c\perp}}{m_{v\perp}m_{v\parallel}m_{c\perp}m_{c\parallel}}\right]^{-1/2}, \qquad \mathcal{F}_L = \left[\frac{\big|m_{v\perp}m_{c\parallel} - m_{v\parallel}m_{c\perp}\big|}{m_{v\perp}m_{v\parallel}m_{c\perp}m_{c\parallel}}\right]^{-1/2}, \tag{4.9}$$

(masses as magnitudes; the sum/difference structure is exactly the $\sigma_c = \mp 1$ dichotomy of (4.5)). GRW's energy-distributed joint density of states (their Eq. (4)) is $D_{l\to u}(E,\hbar\omega) = (8\pi^2\hbar^2)^{-1}\mathcal{F}/k_\parallel$; it differs from the honest measure (4.8) by the constant $K/D_{l\to u} = 32\pi^3\mathcal{F}/\hbar^2$, a normalization GRW absorb into their $|P|^2$ (see App. B); only the fitted strength $S = \mathcal{F}|P|^2_{\mathrm{GRW}}$ is observable.

4.4 Occupation factors

Resolving the $\mathbf{k}$-sum through the final-state energy E (upper-band energy; the initial state then sits at $E - \hbar\omega$), the thermal weights are

$$\text{absorption: } \Delta f = f(E-\hbar\omega) - f(E), \qquad \text{emission: } f(E)\big[1 - f(E-\hbar\omega)\big]. \tag{4.10}$$

For the noble-metal d bands, $E - \hbar\omega \lesssim -1.5\,\mathrm{eV} \ll -k_BT$, so $f(E-\hbar\omega) \simeq 1$ and $\Delta f \to [1 - f(E)]$: GRW's factor. The full form (4.10) is kept because the non-equilibrium extension replaces f by f^S or f^P, for which the lower-band occupation is no longer saturated.

5 The two critical points, side by side

Everything so far is common. The topology now forks the derivation through two signs only: σ_c (shape of $\bar{B}$) and $\mathrm{sign}(\mathcal{D})$ (which window edge is singular). The columns below use the C&S-signed masses of Sec. 4.1; the alternative X branch implied by GRW's own model is treated in Sec. 5.1.

X point $\quad (\sigma_c = -1)$

Bands. Conduction X_6^-: saddle (up along $k_\perp$, down along $k_\parallel$); valence X_7^+: maximum.

$$E_c = E_{0c} + A_c k_\perp^2 - B_c k_\parallel^2,$$
$$E_v = -\hbar\omega_{X_7^+} - A_v k_\perp^2 - B_v k_\parallel^2.$$

Transition surface.

$$\Omega = \mathcal{E}_g^X + \bar{A}_X k_\perp^2 + \bar{B}_X k_\parallel^2, \quad \bar{B}_X = B_v - B_c.$$

With the C&S-signed masses $m_{v\parallel} = 0.15 < m_{c\parallel} = 0.40$: $B_v > B_c$, so $\bar{B}_X > 0$: Ω has a *minimum* at X despite the conduction band being a saddle. CEDS exists only for $\hbar\omega \geq \mathcal{E}_g^X$ and is *closed* (an ellipse in the (u, v) quadrant's boundary sense: a segment).

Determinant.

$$\mathcal{D}_X = A_c B_v + A_v B_c > 0.$$

Window. From $v \geq 0$ with $\mathcal{D}_X > 0$: upper edge; from $u \geq 0$: lower edge:

$$E_X^+ = E_{0c} + \frac{A_c}{\bar{A}_X}(\hbar\omega - \mathcal{E}_g^X),$$
$$E_X^- = E_{0c} - \frac{B_c}{\bar{B}_X}(\hbar\omega - \mathcal{E}_g^X).$$

Slopes for Au: $A_c/\bar{A}_X = 0.38$, $B_c/\bar{B}_X = 0.60$. The window dips *below* E_F once $\hbar\omega > \mathcal{E}_g^X + E_{0c}\bar{B}_X/B_c \simeq 2.22$ eV — the conduction saddle drags final states under the Fermi sea, which is what makes X thermally active.

Singular edge. From (4.4),

$$k_\parallel^2 = \frac{\bar{A}_X}{\mathcal{D}_X}\,(E_X^+ - E),$$

so $K \propto 1/k_\parallel$ diverges at the *upper* limit ($k_\parallel \to 0$ circle of the CEDS):

$$\frac{1}{k_\parallel(E)} = \sqrt{\frac{\mathcal{D}_X}{\bar{A}_X}}\,\frac{1}{\sqrt{E_X^+ - E}}.$$

1D line-shape integral.

$$J_X(\hbar\omega, T) = \int_{E_X^-}^{E_X^+} \frac{\Delta f(E)\,\mathrm{d}E}{\sqrt{E_X^+ - E}}.$$

Onset at $T = 0$: $\hbar\omega = \mathcal{E}_g^X$ exactly, with $J_X \propto \sqrt{\hbar\omega - \mathcal{E}_g^X}$ (M_0-type onset of the closed window).

L point $\quad (\sigma_c = +1)$

Bands. Conduction L_6^-: minimum in both directions; valence L^+ (d top): maximum.

$$E_c = E_{0c} + A_c k_\perp^2 + B_c k_\parallel^2,$$
$$E_v = -\hbar\omega_{L^+} - A_v k_\perp^2 - B_v k_\parallel^2.$$

Transition surface.

$$\Omega = \mathcal{E}_g^L + \bar{A}_L k_\perp^2 + \bar{B}_L k_\parallel^2, \quad \bar{B}_L = B_v + B_c > 0$$

unconditionally: Ω is an M_0 minimum for *any* masses. CEDS exists only for $\hbar\omega \geq \mathcal{E}_g^L$ and is closed (ellipsoid).

Determinant.

$$\mathcal{D}_L = A_c B_v - A_v B_c < 0 \text{ for Au}$$

$(m_{v\perp} m_{c\parallel} = 0.084 < m_{c\perp} m_{v\parallel} = 0.247)$.

Window. Same algebra, but $\mathcal{D}_L < 0$ *reverses both inequalities*: $v \geq 0$ now gives the *lower* edge, $u \geq 0$ the upper:

$$E_L^- = E_{0c} + \frac{A_c}{\bar{A}_L}(\hbar\omega - \mathcal{E}_g^L),$$
$$E_L^+ = E_{0c} + \frac{B_c}{\bar{B}_L}(\hbar\omega - \mathcal{E}_g^L).$$

Slopes for Au: $A_c/\bar{A}_L = 0.745$, $B_c/\bar{B}_L = 0.896$. The whole window rides *upward* from $E_{0c} = 0.89$ eV: in this parameterization the L absorption never touches the Fermi sea for $\hbar\omega \geq \mathcal{E}_g^L$ (see Remark 5.1).

Singular edge. From (4.4) with $\mathcal{D}_L < 0$,

$$k_\parallel^2 = \frac{\bar{A}_L}{|\mathcal{D}_L|}\,(E - E_L^-),$$

so $K \propto 1/k_\parallel$ diverges at the *lower* limit:

$$\frac{1}{k_\parallel(E)} = \sqrt{\frac{|\mathcal{D}_L|}{\bar{A}_L}}\,\frac{1}{\sqrt{E - E_L^-}}.$$

1D line-shape integral.

$$J_L(\hbar\omega, T) = \int_{E_L^-}^{E_L^+} \frac{\Delta f(E)\,\mathrm{d}E}{\sqrt{E - E_L^-}}.$$

Onset at $T = 0$: $\hbar\omega = \mathcal{E}_g^L$ exactly, with $J_L \propto \sqrt{\hbar\omega - \mathcal{E}_g^L}$; the inverse-square-root weight sits at the onset edge itself, which is why the L edge is sharp and strong.

Both points share the M_0-type $\sqrt{\hbar\omega - \mathcal{E}_g}$ JDOS onset; the *only* structural difference in the final integrals is which edge carries the $1/\sqrt{\ }$ singularity — upper at X ($\mathcal{D}_X > 0$), lower at L ($\mathcal{D}_L < 0$). This reconciles the apparent contradiction between the repository documents that call both onsets "M_0" and those that emphasize the X/L asymmetry.

Remark 5.1 (The L level and the Fermi-surface neck) *GRW's qualitative discussion attributes the strength of the L singularity to the onset CEDS being tangent to the Fermi surface along the neck. A neck requires the L_6^- ($L_{2'}$) level* below *E_F, whereas the fitted level scheme ($\mathcal{E}_g^L = 2.45$, $\hbar\omega_{L^+} = 1.56 \Rightarrow E_{0c}^L = +0.89$ eV) places the model window entirely above E_F, making the L channel temperature-inert through the Fermi factor. Both statements cannot hold in one parabolic model; the fitted scheme is what the code implements, and the tension should be flagged in the thesis rather than silently resolved.*

5.1 The X fork: sign of $\bar{B}_X$ (closed window vs. GRW's open CEDS)

The starred X masses in Sec. 4.1 are the crux of a real discrepancy between my C&S extraction and GRW's operational model. Everything above the gap is common; the fork concerns $m_{c\|}$ vs $m_{v\|}$, i.e. the sign of $\bar{B}_X = B_v - B_c$:

Branch A: $\bar{B}_X > 0$ (my extraction)

$m_{c\|} = 0.40$, $m_{v\|} = 0.15$ (d band *steeper* than sp along Δ).

- Closed window $[E_X^-, E_X^+]$ as derived above.
- **No geometric sub-gap transitions**: at $T = 0$, ε_2^X vanishes below $\mathcal{E}_g^X$; the observed tail under 1.94 eV must come entirely from lifetime broadening and thermal smearing.
- GRW's $-20k_BT$ lower cutoff is *not* the true limit; the geometric floor E_X^- lies above it, and integrating below E_X^- adds (numerically tiny at 300 K, growing with T) unphysical weight. The GUI formula sheet currently mixes this branch's masses with Branch B's window — the "chimera" flagged in the repo manifest.

Branch B: $\bar{B}_X < 0$ (GRW's model)

$m_{c\|} = 0.15$, $m_{v\|} = 0.40$ (sp light, d heavy along Δ — the physical prior for flat d bands).

- Ω is a genuine saddle: the CEDS is an *open* hyperboloid for all $\hbar\omega$; transitions exist also for $\hbar\omega < \mathcal{E}_g^X$ (real sub-gap tail).
- No geometric lower limit: the E integral is cut by occupation alone; GRW's practical floor $E_{\min} = -20k_BT$.
- Upper limit is piecewise:

$$\begin{aligned} \hbar\omega \geq \mathcal{E}_g^X &: \quad E^+ = E_{0c} + \tfrac{A_c}{A}(\hbar\omega - \mathcal{E}_g^X), \\ \hbar\omega < \mathcal{E}_g^X &: \quad E^+ = E_{0c} - \tfrac{B_c}{|\bar{B}_X|}(\mathcal{E}_g^X - \hbar\omega), \end{aligned}$$

  singular ($k_\| \to 0$) above the gap; the finite $k_\perp = 0$ edge below it $\Rightarrow$ *step-like onset.* At $T = 0$ absorption starts when E^+ crosses E_F:

$$\hbar\omega_{\text{on}} = \mathcal{E}_g^X - E_{0c}\,\frac{B_c - B_v}{B_c} \simeq 1.83 \text{ eV}$$

  — the piezomodulation threshold.

Remark 5.2 (Fit degeneracy) *Above the gap, the two branches give the* same functional form $\Delta f/\sqrt{E_X^+ - E}$ *with the same E_X^+ (which depends only on perpendicular masses); the parallel masses enter only through $\mathcal{F}_X = \hbar^2/2\sqrt{\mathcal{D}_X}$ ($\mathcal{F}_X^A = 0.170\,m$, $\mathcal{F}_X^B = 0.152\,m$), which is absorbed into the fitted strength $S_X = \mathcal{F}_X|P_X|^2$. A fit to ε_2 above 1.94 eV therefore* cannot *distinguish the branches; they differ below the gap (geometric tail vs broadening-only) and in the extracted $|P_X|^2$.* ***Action****: re-measure the two Δ-axis slopes in C&S Fig. 5 to fix the branch; until then, results* *should quote the branch used.*

6 Assembled ε_2 for gold

Inserting (4.8) and (3.7) into (3.6), with N full critical points,

$$\varepsilon_2^{(i)}(\hbar\omega, T) = \frac{\pi e^2}{\varepsilon_0 m^2\omega^2} \cdot \frac{2}{(2\pi)^3} \cdot \frac{N_i|P_i|^2}{3} \int \mathrm{d}E\, \Delta f(E)\, K_i(E, \hbar\omega) = \frac{e^2 N_i |P_i|^2}{12\pi\, \varepsilon_0 m^2\omega^2\, |\mathcal{D}_i|} \int \frac{\Delta f(E)\,\mathrm{d}E}{k_\parallel(E, \hbar\omega)}, \tag{6.1}$$

i.e., with the explicit kernels of Sec. 5,

$$\boxed{\begin{aligned} \varepsilon_2^X(\hbar\omega, T) &= \frac{e^2\, N_X |P_X|^2}{12\pi\, \varepsilon_0 m^2\omega^2 \sqrt{\bar{A}_X \mathcal{D}_X}} \int_{E_X^-}^{E_X^+} \frac{\big[f(E-\hbar\omega) - f(E)\big]\mathrm{d}E}{\sqrt{E_X^+ - E}}, \\ \varepsilon_2^L(\hbar\omega, T) &= \frac{e^2\, N_L |P_L|^2}{12\pi\, \varepsilon_0 m^2\omega^2 \sqrt{\bar{A}_L |\mathcal{D}_L|}} \int_{E_L^-}^{E_L^+} \frac{\big[f(E-\hbar\omega) - f(E)\big]\mathrm{d}E}{\sqrt{E - E_L^-}}, \end{aligned}} \tag{6.2}$$

with windows and slopes from Sec. 5 (Branch A) or Sec. 5.1 (Branch B at X), and $\varepsilon_2^{\mathrm{inter}} = \varepsilon_2^X + \varepsilon_2^L$. Every constant is now explicit; the dimensional audit is in App. C. Fitted ratio (GRW): $|P_X/P_L|^2 = 0.370$; absolute strengths are set by one overall scale against Johnson & Christy.

Equation (6.2) is equivalent to GRW Eq. (9) $[\varepsilon_2 = \frac{8\pi^2 e_G^2 \hbar^4}{3m^2(\hbar\omega)^2} \sum_i |P_i|^2_{\mathrm{GRW}}\, \mathcal{I}_i]$ under the dictionary of App. B.

7 Photonic density of states and the emission counterpart

7.1 Mode counting

Periodic quantization in volume V: allowed wavevectors $\mathbf{q} = \frac{2\pi}{L}(n_x, n_y, n_z)$, two transverse polarizations, $\omega = cq$:

$$\rho(\omega)\,\mathrm{d}\omega = \frac{2}{V}\frac{V}{(2\pi)^3} 4\pi q^2\,\mathrm{d}q \quad\Longrightarrow\quad \boxed{\rho(\omega) = \frac{\omega^2}{\pi^2 c^3}} \tag{7.1}$$

modes per unit volume per unit angular frequency (per unit photon energy: $\rho(\omega)/\hbar$). In a transparent host of index n, $c \to c/n$ gives $\rho = n^3\omega^2/\pi^2 c^3$ (with dispersion, $n^2(n + \omega\,\mathrm{d}n/\mathrm{d}\omega)$); in a structured environment the role of ρ is played by the *local* density of states $\rho(\mathbf{r}, \omega)$ (Novotny & Hecht), which is precisely the photonic factor of the PL factorization used in this thesis.

7.2 Spontaneous emission rate of one pair

FGR with the quantized field (3.3) at $n_q = 0$, summed over modes:

$$W = \frac{2\pi}{\hbar} \sum_{\mathbf{q},\lambda} \left(\frac{e}{m}\right)^2 \frac{\hbar}{2\varepsilon_0 \omega_q V} \big|\hat{\varepsilon}_\lambda \cdot \mathbf{p}_{cv}\big|^2 \delta(\Omega - \hbar\omega_q). \tag{7.2}$$

Using $\sum_\lambda \int \mathrm{d}\Omega_q |\hat{\varepsilon}_\lambda \cdot \mathbf{p}_{cv}|^2 = \frac{8\pi}{3}|\mathbf{p}_{cv}|^2$ and $\sum_{\mathbf{q}} \to V\int q^2 \mathrm{d}q\, \mathrm{d}\Omega_q/(2\pi)^3$,

$$\boxed{W_{\mathrm{sp}}(\omega) = \frac{e^2\,\omega\,|P|^2}{3\pi\varepsilon_0 \hbar m^2 c^3} = \underbrace{\frac{\pi e^2 |P|^2}{3\varepsilon_0 \hbar m^2 \omega}}_{\text{matter}} \times \underbrace{\frac{\omega^2}{\pi^2 c^3}}_{\rho(\omega)}} \tag{7.3}$$

— the vacuum Einstein-A rate in the $\mathbf{A}\cdot\mathbf{p}$ form (equivalently $e^2\omega^3|\langle\mathbf{r}\rangle_{cv}|^2/3\pi\varepsilon_0\hbar c^3$, via $|\mathbf{p}_{cv}| = m\omega|\langle\mathbf{r}\rangle_{cv}|$). The factorization into a matter part times $\rho(\omega)$ is exact and is the microscopic origin of the LDOS $\times$ electronic split assumed in Chapter 2.

7.3 The interband PL spectrum and the Kirchhoff check

Summing (7.3) over occupied-upper/empty-lower pairs of one critical point (spin 2, N points):

$$\boxed{R(\hbar\omega) = \rho(\omega)\,\frac{\pi e^2}{3\varepsilon_0\hbar m^2\omega}\,\frac{N|P|^2}{4\pi^3}\int \mathrm{d}E\; f(E)\big[1 - f(E-\hbar\omega)\big]\,K(E,\hbar\omega)} \tag{7.4}$$

photons per unit time, volume and photon energy, with the same kernels and windows as in (6.2) (only the occupation factor differs). Using the thermal identity (notes A2/A3)

$$f(E)\big[1 - f(E-\hbar\omega)\big] = n_B(\hbar\omega)\,\big[f(E-\hbar\omega) - f(E)\big], \qquad n_B(x) = \frac{1}{\mathrm{e}^{x/k_BT} - 1}, \tag{7.5}$$

every electronic factor of (7.4) maps onto (6.2) and the constants collapse *exactly*:

$$\boxed{R(\hbar\omega) = \frac{\omega^3}{\pi^2\hbar c^3}\,\varepsilon_2(\omega)\,n_B(\hbar\omega) = \frac{\omega\,\rho(\omega)}{\hbar}\,\varepsilon_2(\omega)\,n_B(\hbar\omega)} \tag{7.6}$$

— the van Roosbroeck–Shockley relation. Emission over (stimulated) absorption per thermal photon is $n_B/(n_B{+}1) = \mathrm{e}^{-\hbar\omega/k_BT}$: Kirchhoff's law, consistent with note A9 and with the selective (non-Planckian) emissivity found for gold. That (6.2) and (7.4), derived independently, satisfy (7.6) with no leftover constant is the strongest global check that all prefactors in this document are right.

Out of equilibrium, replace f in (7.4) by f^S (CW) or f^P (pulsed); (7.6) then no longer holds and the deviation *is* the non-equilibrium PL signal this thesis is after. In a structured photonic environment, $\rho(\omega) \to \rho(\mathbf{r},\omega)$ in (7.4).

A Corrections to GRW's printed Eqs. (6) and (8)

Eq. (6). As printed (and as OCR'd), the right-hand side is dimensionally inconsistent (an energy under a square root equated to a wavevector). Solving (4.3) for $v = k_\parallel^2$ in GRW's variables gives the intended equation:

$$k_\parallel^2 = \frac{2\mathcal{F}_X^2}{\hbar^2}\Big[\frac{\hbar\omega - \hbar\omega_{X_7^+} - E}{m_{c\perp}} + \frac{\hbar\omega_{X_6^-} - E}{m_{v\perp}}\Big] = \frac{A_c(\hbar\omega - \mathcal{E}_g^X) - \bar{A}_X(E - E_{0c})}{\mathcal{D}_X}, \tag{6$'$}$$

whose three numerator terms match the printed fragments slot by slot (the print lost the $1/m_{c\perp}$ on the first term, the global $2\mathcal{F}^2/\hbar^2$, and the square on $k_\parallel$).

Eq. (8). Printed: $E_{\max} = \hbar\omega_{X_6^-} + (\mathcal{E}_g^X - \hbar\omega)\,m_{u\parallel}/(m_{l\parallel} - m_{u\parallel})$. Solving the model (4.1) for the $k_\perp = 0$ edge gives

$$E_{k_\perp=0} = \hbar\omega_{X_6^-} + (\mathcal{E}_g^X - \hbar\omega)\,\frac{m_{l\parallel}}{m_{u\parallel} - m_{l\parallel}}, \tag{8$'$}$$

i.e. the printed equation has the subscripts $u\parallel$ and $l\parallel$ interchanged (a single consistent swap; Eq. (6)'s subscripts are *not* swapped, so this is specific to (8)). On GRW's Branch B ($m_{u\parallel} = 0.15 < m_{l\parallel} = 0.40$), (8′) is the finite $k_\perp{=}0$ top edge of the *sub-gap* CEDS — the correct upper integration limit for $\hbar\omega < \mathcal{E}_g^X$ and the origin of the step-like 1.83 eV onset. On Branch A ($\bar{B}_X > 0$), the same algebraic edge is instead the *lower* window limit E_X^-; the earlier repo diagnosis (`rosei_equivalence_derivation.tex`: "(8) should contain perpendicular masses") implicitly assumed Branch A above the gap, where the binding upper edge is indeed the $k_\perp$-mass expression E_X^+. The branch-resolved statement above supersedes both.

B Dictionary of conventions across the repository

Document	upper band	lower band	$\bar{B}$	status
GRW 1975; *X/L note*; this doc (X)	$+A_u k_\perp^2 - B_u k_\parallel^2$	$-A_l k_\perp^2 - B_l k_\parallel^2$	$B_l - B_u$	canonical
Equivalence note (X)	$+A_u k_\perp^2 + B_u k_\parallel^2$	$-A_l k_\perp^2 + B_l k_\parallel^2$	$B_u - B_l$	same $\mathcal{D}$ (sign-robust)
A8 / A8.L	$+A_c u - B_c v$	$-A_v u - B_v v$	$B_v - B_c$	A8 ok[(a)]; A8.L wrong at L[(b)]
`interband_derivation_v2`	mixed	mixed	—	internally inconsistent[(c)]

[(a)] A8's $\mathcal{E}_{\min}$ denominator ($B_v{+}B_c$) is the $\sigma_c{=}{+}1$ formula; for its own X conventions it should be $\bar{B} = B_v{-}B_c$. [(b)] A8.L models the L conduction band as a saddle ($-B_c$); C&S give $m^L_{c\parallel} = +0.12$: it is a minimum, $\sigma_c = +1$. [(c)] v2 asserts $\bar{B} > 0$ *and* a geometric sub-gap tail (requires $\bar{B} < 0$), and labels L as M_1.

EDJDOS normalizations. With K from (4.8) (this doc), $\mathcal{J}$ from the *Equivalence* note ($k_\parallel > 0$ half only: $\mathcal{J} = K/2$), and GRW's $D = (8\pi^2\hbar^2)^{-1}\mathcal{F}/k_\parallel$:

$$K = 2\,\mathcal{J} = \frac{32\pi^3\mathcal{F}}{\hbar^2}\,D. \tag{B.1}$$

Matching (6.2) to GRW Eq. (9) fixes GRW's matrix-element convention:

$$|P|^2_{\mathrm{GRW}} = \frac{4N\mathcal{F}}{\hbar^4}\,|P|^2_{\mathrm{SI}} = \frac{N}{\mathcal{F}\,\mathcal{D}}\,|P|^2_{\mathrm{SI}}, \tag{B.2}$$

which carries dimension $[p^2]\,[\text{mass}]\,[\hbar]^{-4}$: GRW's $|P|^2$ is not a bare momentum-squared, and only the combination $S = \mathcal{F}|P|^2_{\mathrm{GRW}}$ (their Eq. (10)) is observable — consistent with their choice to fit S_X, S_L rather than $|P|^2$ alone.

C Dimensional audit

All checks in SI; $[\,e^2/\varepsilon_0\,] = \mathrm{J\,m}$, $[\,|P|^2\,] = \mathrm{kg^2m^2s^{-2}}$, $[\,A_i\,] = [\,B_i\,] = \mathrm{J\,m^2}$, $[\,\mathcal{D}\,] = \mathrm{J^2m^4}$, $[\,\mathcal{F}\,] = \mathrm{kg}$.

- (4.8): $[K] = 1/([\mathrm{J^2m^4}][\mathrm{m^{-1}}]) = \mathrm{J^{-2}m^{-3}}$ — states per volume per (energy)^2, as a double-δ measure must be.
- (6.2): $[\bar{A}\mathcal{D}]^{1/2} = \mathrm{J^{3/2}m^3}$; the integral gives $\mathrm{J^{1/2}}$; $\dfrac{\mathrm{J\,m\cdot kg^2m^2s^{-2}\cdot J^{1/2}}}{\mathrm{kg^2s^{-2}\cdot J^{3/2}m^3}} = 1.$ ✓
- (7.1): $[\rho] = \mathrm{s\,m^{-3}}$ (modes per volume per dω). ✓
- (7.3): $\dfrac{\mathrm{J\,m\cdot s^{-1}\cdot kg^2m^2s^{-2}}}{\mathrm{J\,s\cdot kg^2\cdot m^3s^{-3}}} = \mathrm{s^{-1}}$. ✓
- (7.4): $[W_{\mathrm{sp}}]\cdot[K]\cdot\mathrm{J} = \mathrm{s^{-1}J^{-1}m^{-3}}$ — photons per time, volume and photon energy. ✓
- (7.6): $[\omega\rho/\hbar] = \mathrm{s^{-1}\,s\,m^{-3}\,J^{-1}} = \mathrm{J^{-1}m^{-3}s^{-1}}$, times dimensionless $\varepsilon_2 n_B$. ✓

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- Vault notes consolidated here: A1–A4, A8, A8.L, *Absorption Integral (Using Rosei's Notations)*, *Interband Absorption at X and L (Rosei's Notation)*, *Equivalence of My Absorption Integral and Rosei's*, *Effective Mass Extraction*, `rosei_model_formulas.md`.